



Histamine and Antihistamines

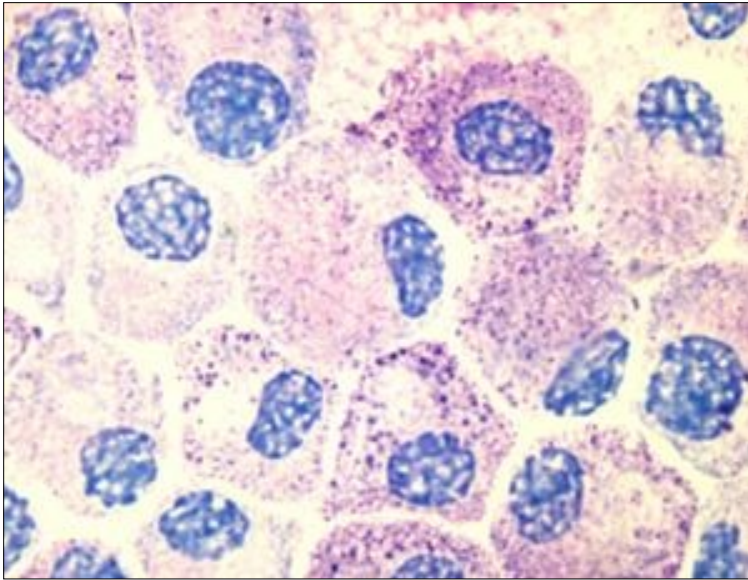
By Yitagesu T (MSc.)



What are allergies?

- Allergies are caused by a hypersensitivity reaction of the **antibody** class **IgE** (which are located on **mast cells** in the tissues and **basophils** in the blood)
- When an allergen is encountered, it binds to **IgE**, which excessively activates the **mast cells** or **basophils**, leading them to release massive amounts of histamines.
- These histamines lead to inflammatory responses ranging from **runny nose** to **anaphylactic shock**
- If both parents have allergies, you have a 70% of having them, if only one parent does, you have a 48% chance (American Academy of Asthma, Allergies and Immunology, Spring 2003).

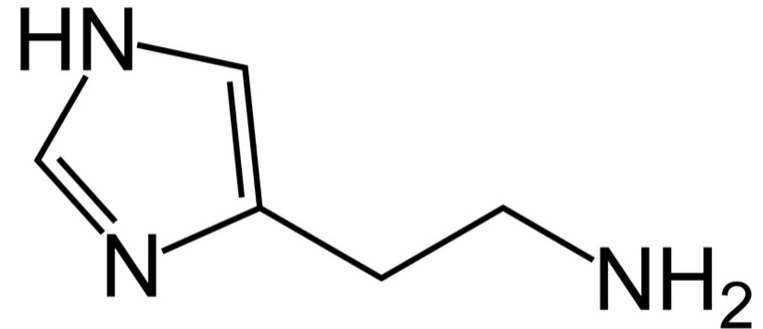
Histamine



Mast Cells

- Histamine is distributed in Mast Cells in all peripheral tissues of the body and **basophils**, which circulate in the blood

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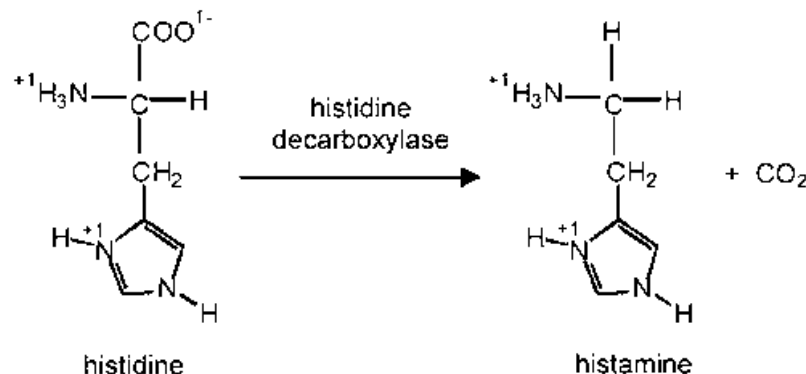


Structure of Histamine

- When it is released, histamine causes inflammation by increasing **vasodilation, capillary permeability, causing smooth muscle contraction, mucus secretion, and parasympathetic nerve stimulation**

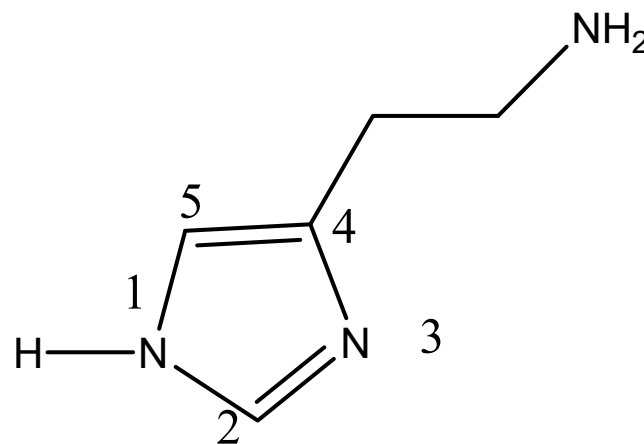
Synthesis of Histamine

- Formed from the amino acid **Histidine** in a decarboxylation reaction with the enzyme **histidine decarboxylase**
- Occurs primarily in **mast cells** and **basophils**

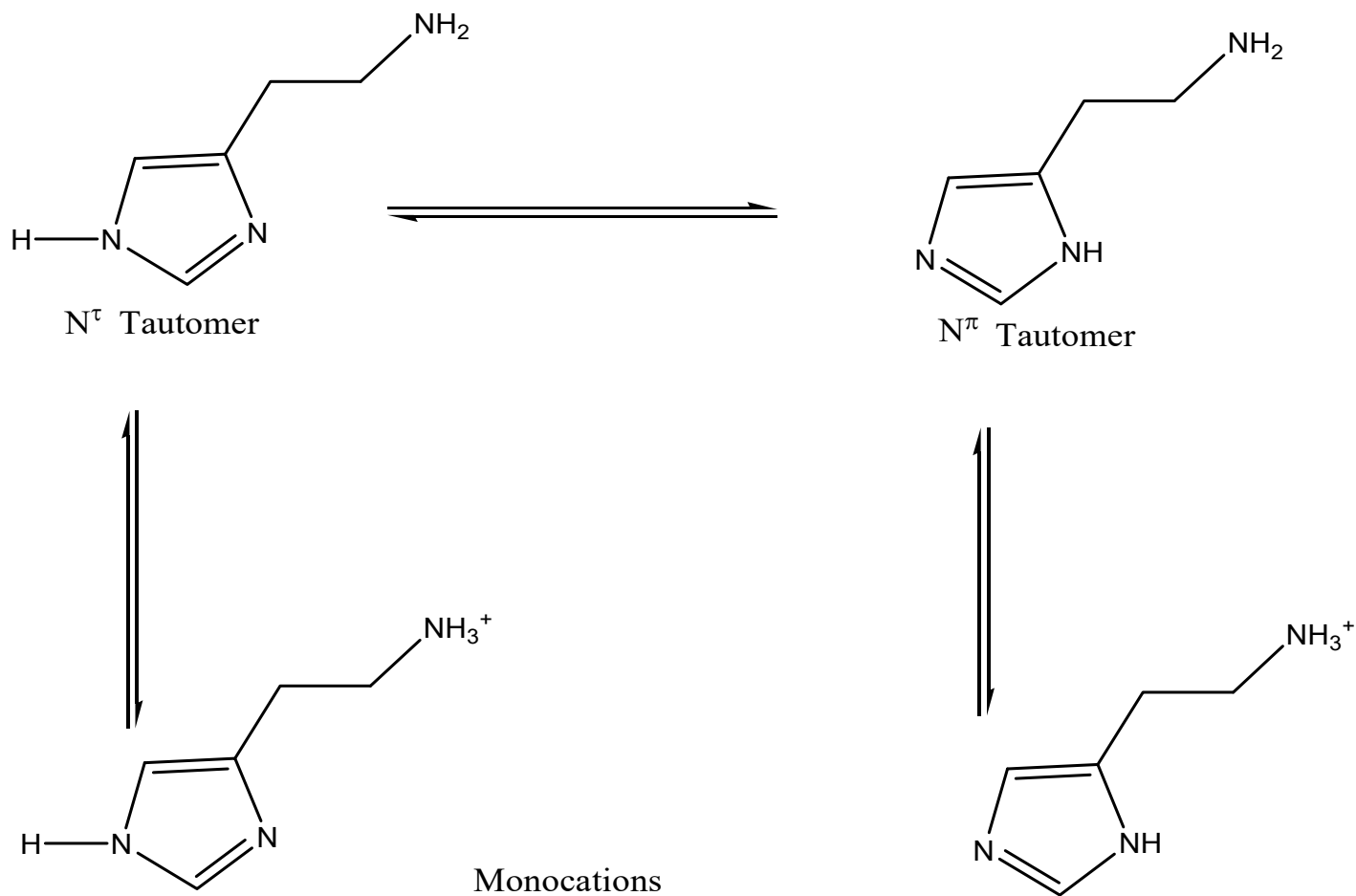


Chemistry of Histamine

- The methylene group of the amino ethyl side chain are named as α -and β .
- The side chain is attached, via the β -CH₂ group to the 4-position of an imidazole ring.



Ionization and tautomerism



Conditions that Release Histamine

- **Tissue injury:**
 - Any physical or chemical agent that injures tissue, skin or mucosa are particularly sensitive to injury and will cause the immediate release of histamine from mast cells.
- **Allergic reactions:**
 - exposure of an antigen to a previously sensitized (exposed) subject can immediately trigger allergic reactions.
- **Drugs and other foreign compounds:**
 - morphine, dextran, antimalarial drugs, dyes, antibiotic bases, alkaloids, amides, quaternary ammonium compounds, enzymes (phospholipase C).
 - Penicillins, Tetracyclines,
 - Basic drugs- amides, amidines, diamidines,
 - Toxins, venoms,
 - Proteolytic enzymes,
 - Bradykinin, Kallidin, & Substance P

Types of Histamine Receptors

- **H1 receptors:**
 - Mediate effects on smooth muscle leading to
 - Vasodilation
 - Increased vascular permeability
 - Contraction of nonvascular smooth muscle
- **H2 receptors:**
 - Mediate histamine stimulation of **gastric acid secretion** and may be involved in **cardiac stimulation**.
- **H3 receptors:**
 - modulator of neurotransmitter release in the central and peripheral nervous systems

What is an antihistamine?

- A drug that reduces or eliminates the effects mediated by the chemical histamine
- Histamine is released by your body during an **allergic reaction** and **acts on a specific histamine receptor**
- Antihistamines compete with histamine for binding sites at the receptors.
- Antihistamine cannot remove the histamine if it is already bound

Clinical Uses of Antihistamines

- Allergic rhinitis (common cold)
- Allergic conjunctivitis (pink eye)
- Allergic dermatological conditions
- Urticaria (hives)
- Angioedema (swelling of the skin)
- Pruritus (atopic dermatitis, insect bites)
- Anaphylactic reactions (severe allergies)
- Nausea and vomiting (first generation H₁-antihistamines)
- Sedation (first generation H₁-antihistamines)

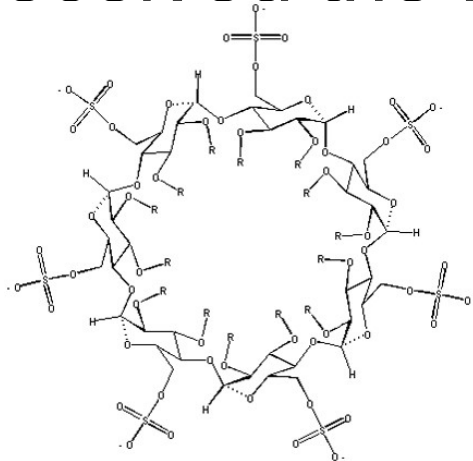


Adverse side effects

- Associated with the first generation H₁-antihistamines and due to their lack of selectivity for the **H₁ receptor** and are muscarinic acetylcholine receptor antagonists (anticholinergic agents), and may also have action at alpha-adrenergic receptors and 5-HT receptors
- Sedation, Dizziness, Tinnitus (ringing in the ear), Blurred vision, Euphoria, Uncoordination, Anxiety, Insomnia, Tremor, Nausea/vomitting, Dry mouth/dry cough
- **Newer second generation H₁-antihistamines** are more selective for the peripheral histamine receptors and have far less side effects (drowsiness, fatigue, headache, nausea and dry mouth)

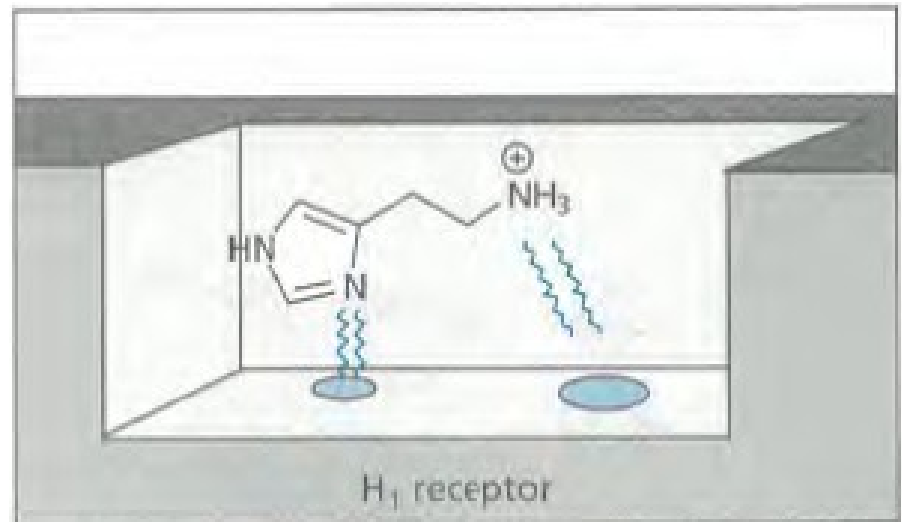
First antihistamine

- Piperoxan
- Discovered in 1933 by **Jeff Forneau** and **Daniel Bovent** while developing a guinea pig animal model of anaphylaxis
- They received the Nobel Prize in 1957



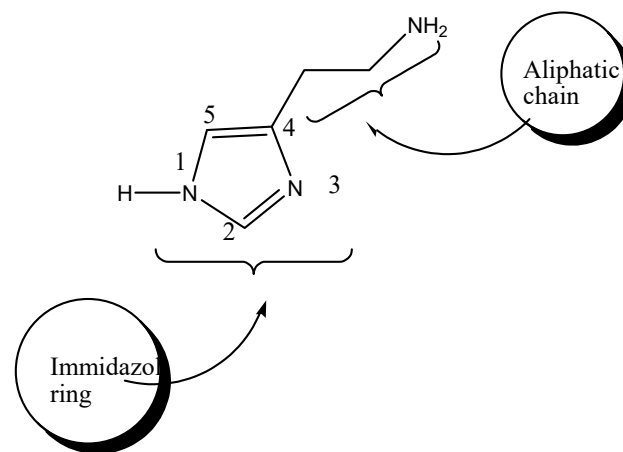
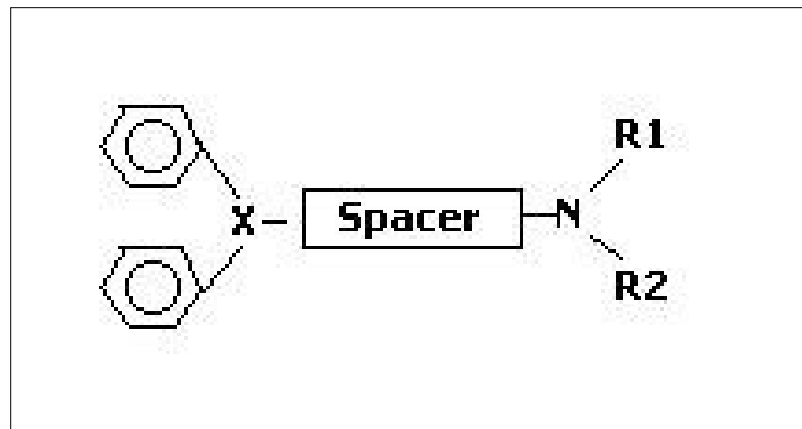
Classes of first generation H₁ receptor antagonist antihistamines

- Ethylenediamines
- Ethanolamines
- Alkylamines
- Piperazines
- Tricyclics



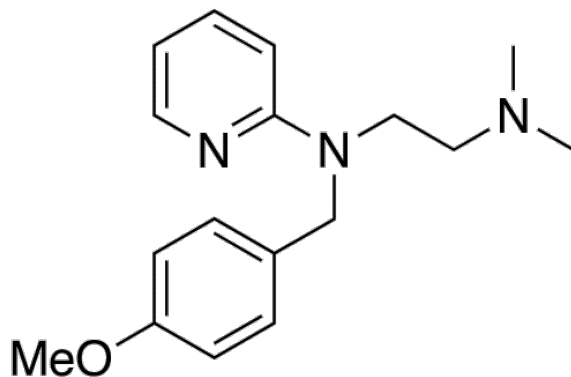
Common Structural Features of classical first generation antihistamines

- 2 aromatic rings, connected to a central carbon, nitrogen, or oxygen
- Spacer between central atom and the amine, usually 2-3 carbons in length. (Can be linear, ring, branched, saturated or unsaturated)
- The amine is substituted with small alkyl groups
- Chirality at X and having the rings in different planes increases potency of the drug



Ethylenediamines

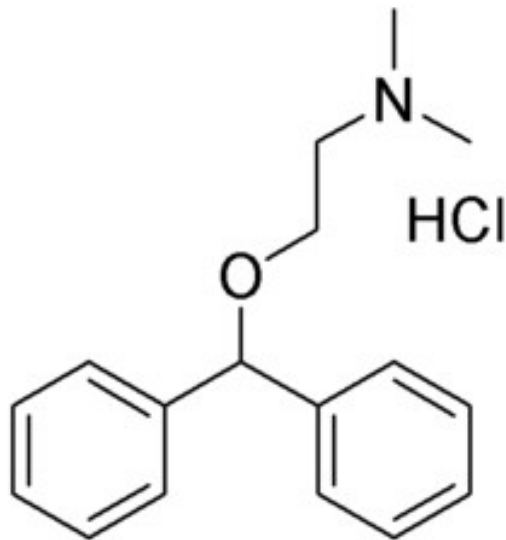
- These were the first group of clinically effective H₁-antihistamines



Mepyramine (Pyrilamine)

Ethanolamines

- This class has significant anticholinergic side effects and sedation, however reduced the gastrointestinal side effects

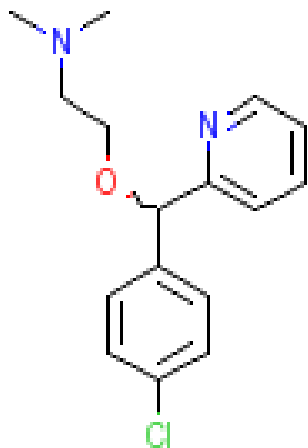


Diphenhydramine (Benedryl)

- Oldest and most effective antihistamine on the market
- Available over the counter
- Because it induces sedation, it's used in nonprescription sleep aids
- Also inhibits the reuptake of serotonin, which led to the search for viable antidepressants with similar structures (prozac)

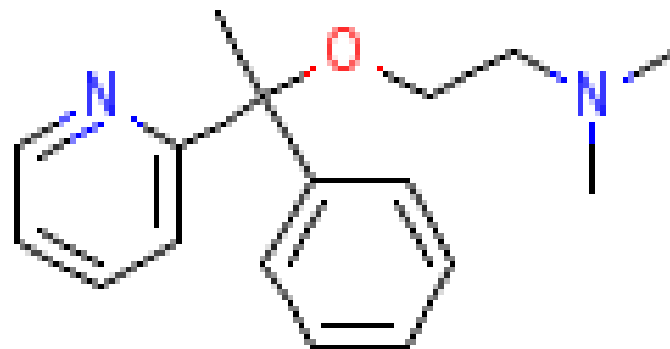
Ethanolamines

Carbinoxamine(Clistine)



- Is used to treat Hay fever and is especially popular to children due its its mild taste
- After 21 reported deaths in children under 2, its now only marketed to children above 3 (FDA, June 2006)

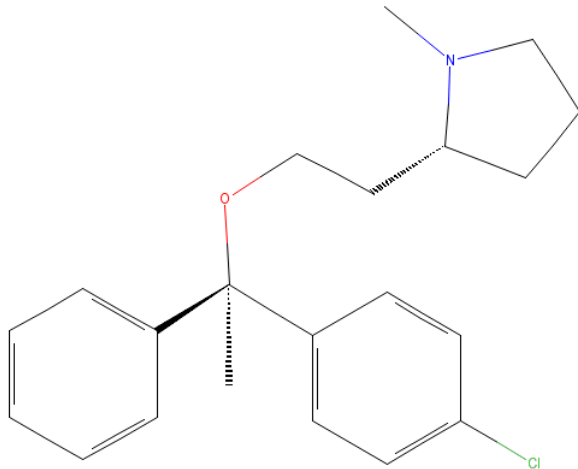
Doxylamine succinate



- Has potent anti-cholinergic effects

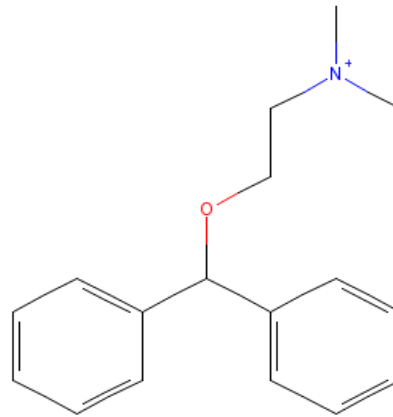
Ethanolamines

Clemastine (Tavist)

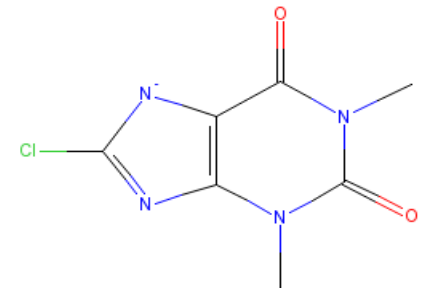


- Exhibits fewer side effects than most antihistamines
- Widely used as an antipruritic (stops itching)

Dimenhydrinate (Dramamine)



- Anti-emetic (anti nausea)
- Also causes strong sedation
- diphenhydramine (Benadryl) and 8-chlorotheophyllinate
- Readily crosses the BBB

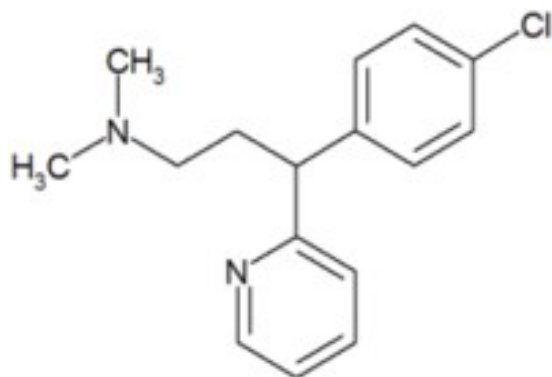


Alkylamines

- Isomerism is an important factor in this class of drugs, which is due to the positioning and fit of the molecules in the H1-receptor binding site
- These drugs have fewer sedative and GI adverse effects, but a greater incidence of CNS stimulation
- These drugs lack the “spacer molecule” (which is usually a nitrogen or oxygen) between the two aromatic rings and at least one of the rings has nitrogen included in the aromatic system

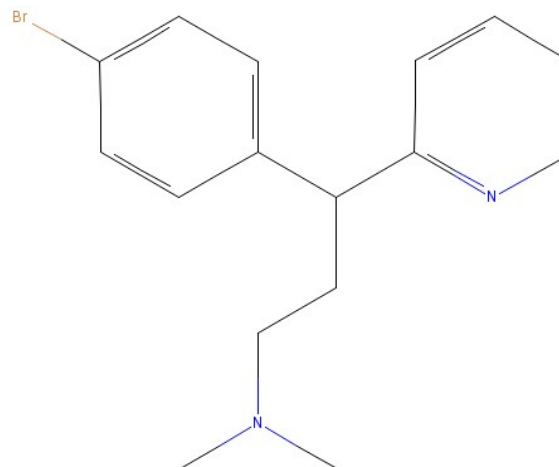
Alkyl amines

Chlorphenamine



- Originally used to prevent allergic conditions
- Shown to have antidepressant properties and inhibit the reuptake of serotonin

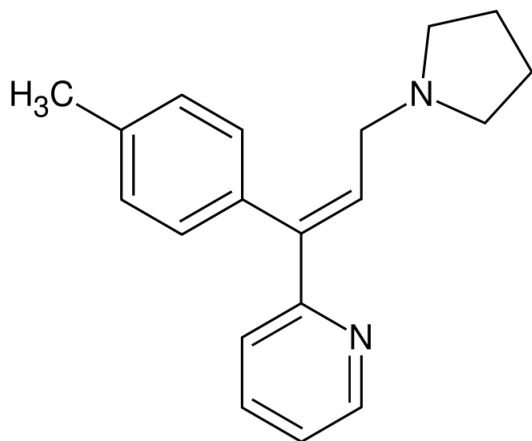
Brompheniramine (Dimetapp)



- Available over the counter
- Used to treat the common cold by relieving runny nose, itchy, watery eyes and sneezing

Alkylamines

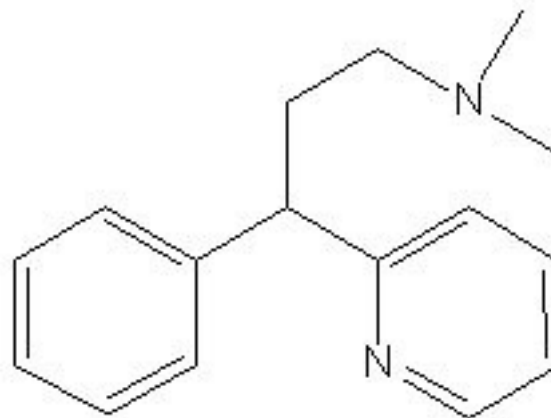
Triprolidine hydrochloride



- Used to alleviate the symptoms associated with allergies
- Can be combined with other cold medicine to relieve “flu-like” symptoms

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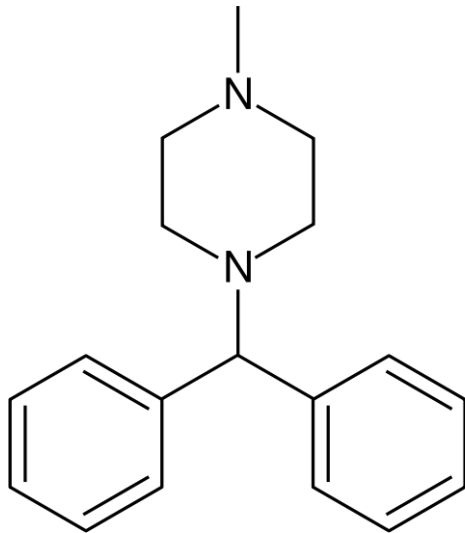
Pheniramine (Avil)



- Used most often to treat hay fever or urticaria (hives)

Piperazines

- Structurally related to the ethylenediamines and the ethanolamines and thus produce significant anti-cholinergic effects
- Used most often to treat motion sickness, vertigo, nausea and vomiting

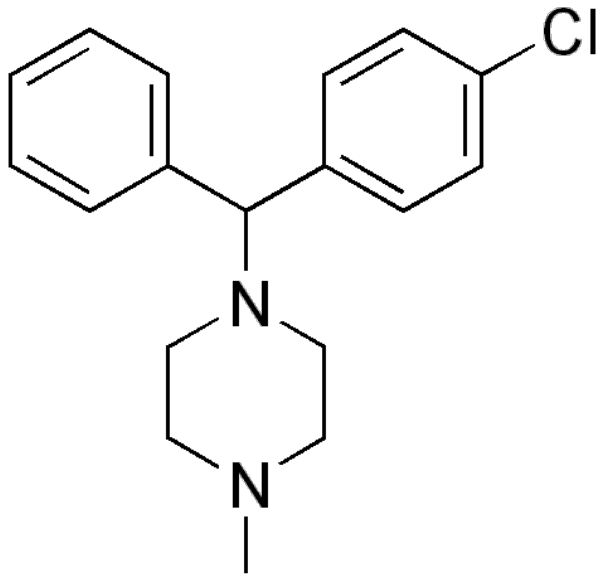


Cyclizine

- Used to treat the symptoms associated with **motion sickness, vertigo and post-op following administration of general anaesthesia and opioids**
- Mechanism of inhibiting motion sickness is not well understood, but it may act on the labyrinthine apparatus and the **chemoreceptor trigger zone** (area of the brain which receives input and induces vomiting)

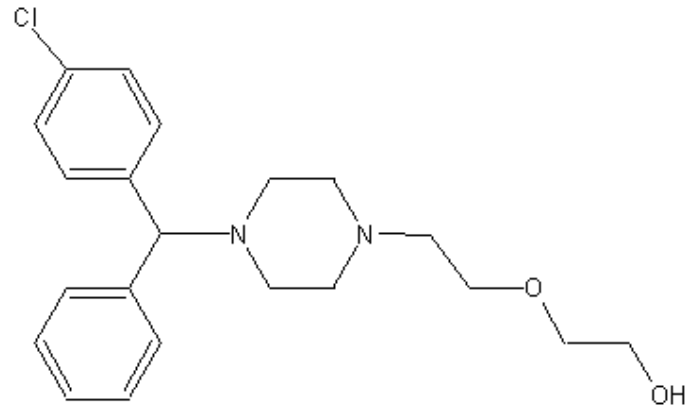
Piperazines

Chlorcyclizine



- This drug is used to treat motion sickness

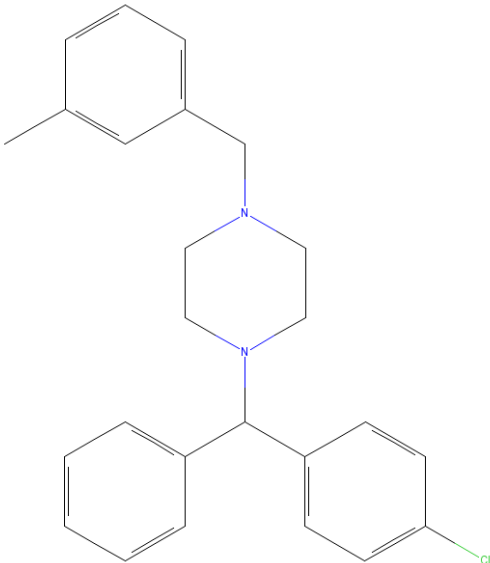
Hydroxyzine



- In addition to treating itches and irritations, its an anitemetic, a weak analgesic and an anxiolytic (treat anxiety)

Piperazines

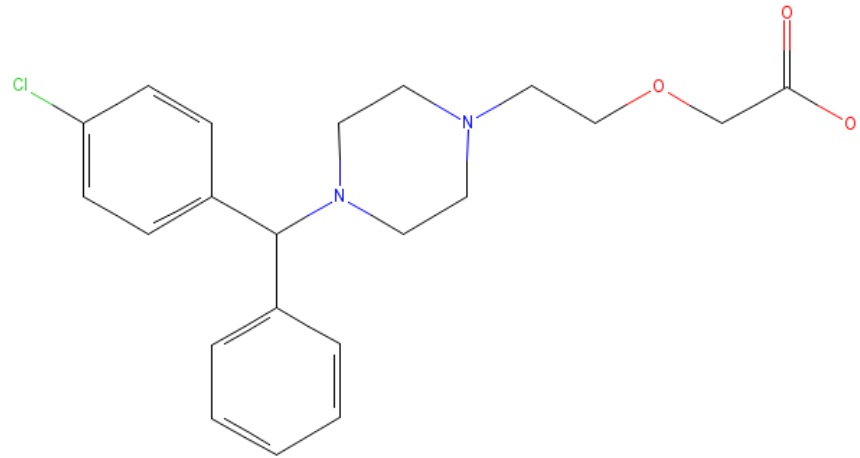
Meclizine



- It is most commonly used to inhibit nausea and vomiting as well as vertigo, however it does cause drowsiness

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Cetirizine (Zyrtec)

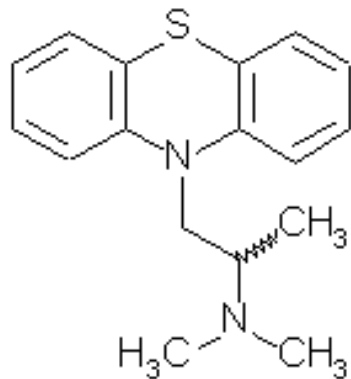


- This drug treats indoor and outdoor allergies and is safe to use in children as young as 2

Tricyclics

- These drugs are structurally related to tricyclic antidepressants, which explains why they have cholinergic side effects

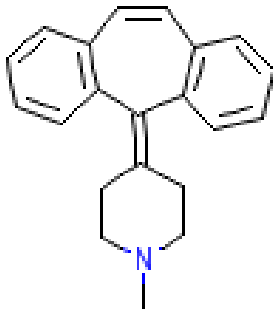
Promethazine (Phenegan)



- This drug has extremely strong anticholinergic and sedative effects
- It was originally used as an antipsychotic, however now it is most commonly used as a sedative or antinausea drug (also severe morning sickness) and requires a prescription

Tricyclics

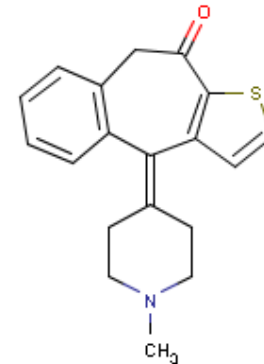
Cyproheptadine



- This drug both an antihistamine and an antiserotonergic agent
- It is a 5-HT₂ receptor antagonist and also blocks calcium channels
- Used to treat hay fever and also to stimulate appetite in people with anorexia
- It is also rarely used to treat Cushing's Syndrome (high level of cortisol in the blood) and migraine headaches

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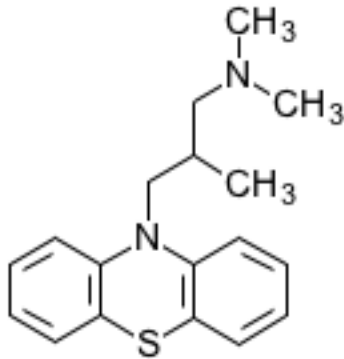
Ketotifen (Zaditor)



- This drug is available in two forms: an **ophthalmic form used to treat allergic conjunctivitis or itchy red eyes** and an **oral form used to prevent asthma attacks**
- It has several adverse side effects including drowsiness, weight gain, dry mouth, irritability and increased nosebleeds

Tricyclics

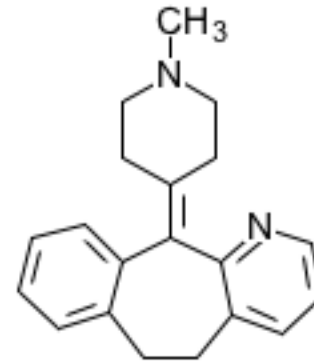
Alimemazine (Vallergan)



- This drug is used to treat itchiness and hives that results from allergies
- Since it causes drowsiness, it is useful for rashes that itch worse at night time
- It is also used to sedate young children before operations

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Azatadine (Optimine or Trinalin)



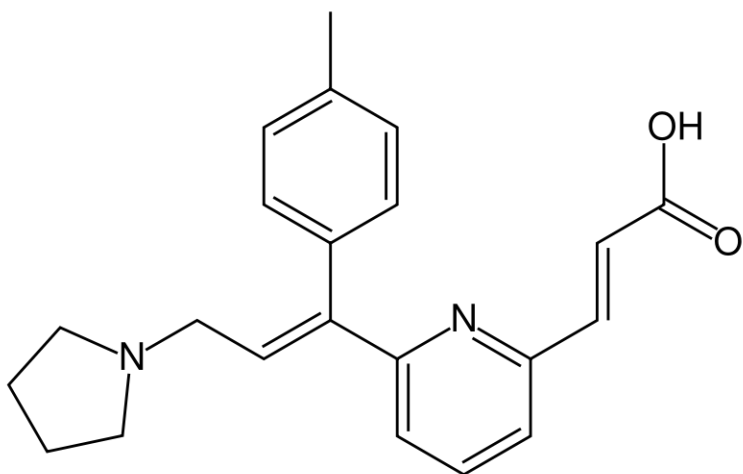
- This drug is used to treat symptoms of allergies and the common cold such as sneezing, runny nose, itchy watery eyes, itching, hives and rashes

Second generation H₁-receptor antagonists

- These are the newer drugs and they are much more **selective for the peripheral H₁-receptors involved in allergies as opposed to the H₁-receptors in the CNS**
- Therefore, these drugs provide the same relief with many fewer adverse side effects
- The structure of these drugs varies and there are no common structural features associated with them
- They are however **bulkier** and **less lipophilic** than the first generation drugs, therefore they do not cross the BBB as readily
- Recent studies have also showed that these drugs also **have anti-inflammatory activity** and therefore, would be helpful in the management of inflammation in allergic airways disease (Devalia and Davies).

Second generation H₁-receptor antagonists

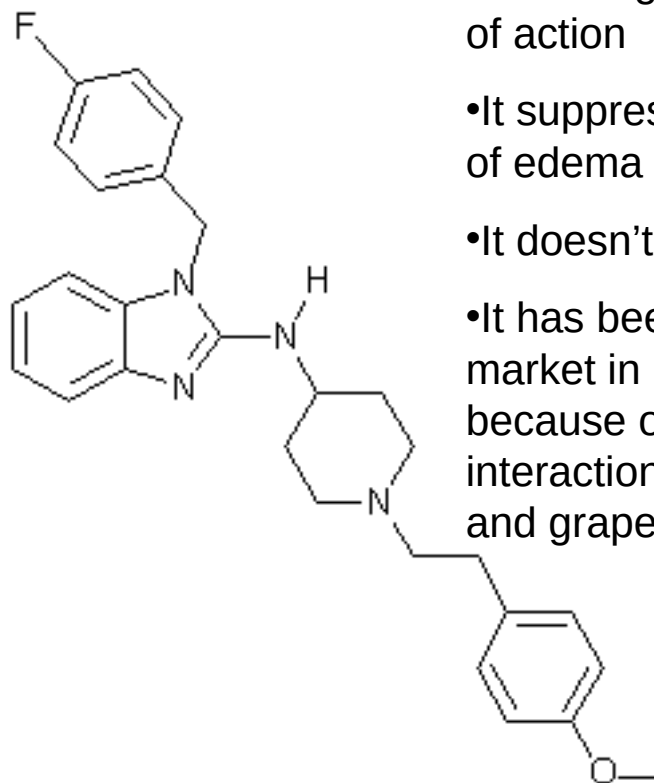
Acrivastine (Semprex-D)



- This drug relieves itchy rashes and hives
- It is non-sedating because it does not cross the BBB

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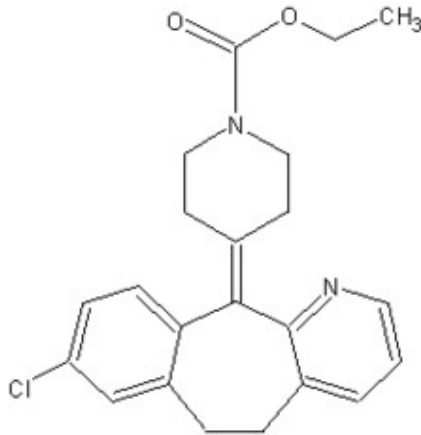
Astemizole (Hismantol)



- This drug has a long duration of action
- It suppresses the formation of edema and puritus
- It doesn't cross the BBB
- It has been taken off the market in most countries because of adverse interactions with erythromycin and grapefruit juice

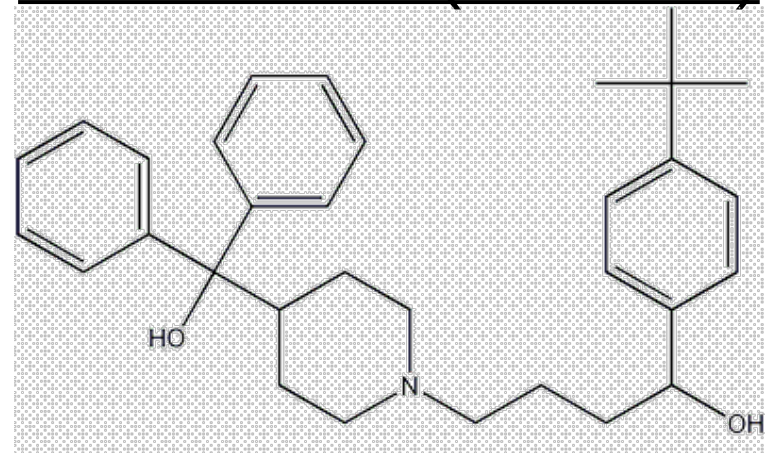
Second generation H₁-receptor antagonists

Loratadine (Claritin)



- It is the only drug of its class available over the counter
- It has long lasting effects and does not cause drowsiness because it does not cross the BBB

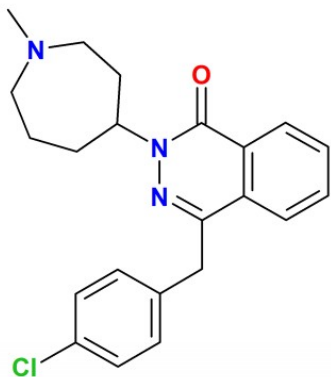
Terfenadine (Seldane)



- It was formerly used to treat allergic conditions
- In the 1990's it was removed from the market due to the increased risk of cardiac arrhythmias

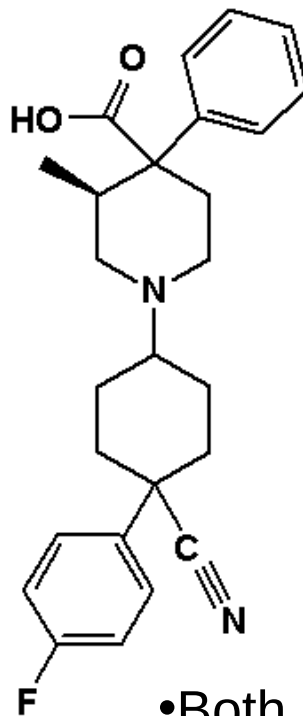
Second generation H₁-receptor antagonists

Azelastine (Astelin or Optivar)



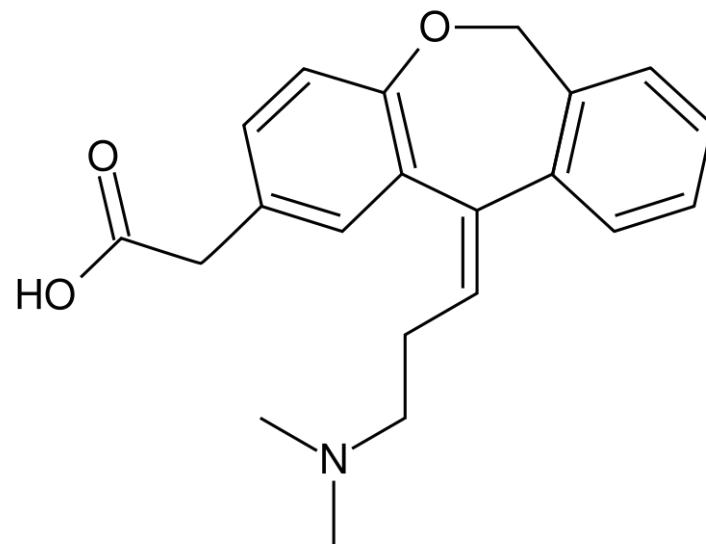
- It is a mast cell stabilizer
- Available as a nasal spray (Astelin) or eye drops for pink eye (Optivar)

Levocabastine (Livostin)



- Both of these drugs are used as eye drops to treat allergic conjunctivitis

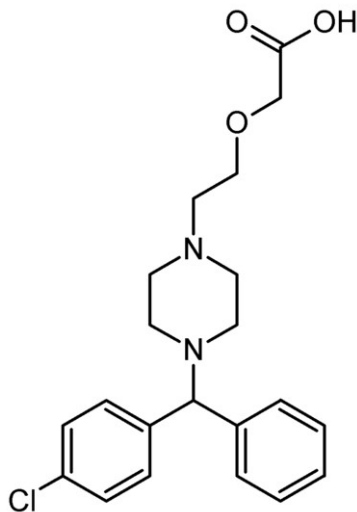
Olopatadine (Patanol)



Third generation H₁-receptor antagonists

- These drugs are derived from second generation antihistamines
- They are either the active enantiomer or metabolite of the second generation drug designed to have increased efficacy and fewer side effects

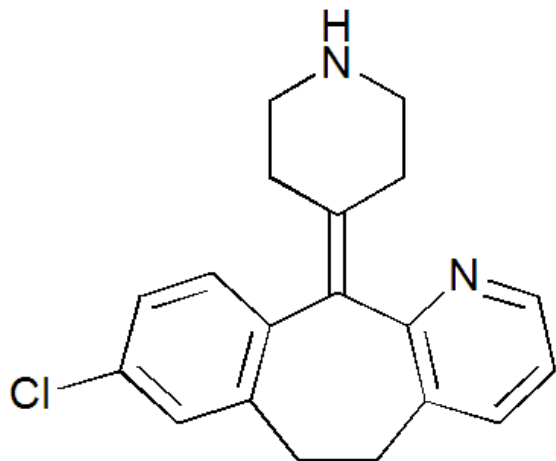
Levocetirizine (Zyzal)



- This drug is the active enantiomer of cetirizine and is believed to be more effective and have fewer adverse side effects.
- Also it is not metabolized and is likely to be safer than other drugs due to a lack of possible drug interactions (Tillement).
- It does not cross the BBB and does not cause significant drowsiness
- It has been shown to reduce asthma attacks by 70% in children

Third generation H₁-receptor antagonists

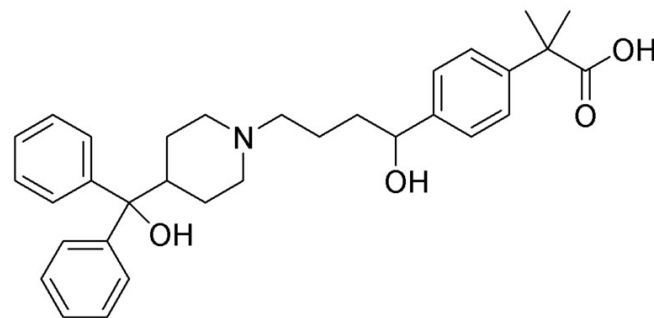
Deslortadine (Clarinet)



- **It is the active metabolite of Lortadine**

- Even though it is thought to be more effective, there is no concrete evidence to prove this

Fexofenadine (Allegra)



- It was developed as an alternative to Terfenadine
- Fexofenadine was proven to be more effective and safe

The future of antihistamines

- The anti-inflammatory activity of second generation antihistamines, about which little is known, will continue to be researched and possibly lead to an effective alternative to corticosteroids in the treatment of allergic airways conditions.
- The action of the H4 receptor will also continue to be researched and will possibly lead to effective treatment of autoimmune diseases.
- Creating antihistamines with higher selectivity and less adverse side effects will continue to be the goal.